Trends in Neurosciences

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Review

Making Dopamine Connections in Adolescence

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A dramatic maturational process ongoing in adolescence is prefrontal cortex development, including its dopamine innervation. Dopamine axons grow from the striatum to the prefrontal cortex, the only known case of long-distance axon growth during adolescence. This is coordinated by the Netrin-1 guidance cue receptor DCC (deleted in colorectal cancer), which in turn controls the intrinsic development of the prefrontal cortex itself. Stimulant drugs in adolescence alter DCC in dopamine neurons and, in turn prefrontal cortex maturation, impacting cognitive abilities. Variations in DCC expression are linked to psychiatric conditions of prefrontal cortex dysfunction, and microRNA regulation of DCC may be key to determining adolescent vulnerability or resilience. Since early interventions are proving to effectively ameliorate disease outcome, the Netrin-1 system is a promising therapeutic target.

Adolescence as a Period of Behavioural and Cognitive Transition

Adolescence is the time when individuals must negotiate the transition between childhood and adulthood at several levels (molecular, cellular, physical, and social). This transition requires a diverse knowledge and skill set pertaining to independent living and the ability to navigate emotional, social, and psychological stimuli far more varied and complex than those experienced during childhood. As such, adolescence is characterised by dramatic behavioural and cognitive changes.

Concurrent with and underlying these adolescent changes is ongoing brain development, in particular at the level of the **prefrontal cortex** (PFC; see Glossary). This region, heavily involved in the complex cognitive and social processes that mature during adolescence, continues to develop into early adulthood (e.g., [1]). One of the key changes to the PFC during adolescence is that its dopamine input continues to increase [2–6]. This dopamine growth in the PFC is unique; maximum innervation levels of other neuromodulators are typically reached before the onset of adolescence [7]. Furthermore, protracted dopamine innervation to the PFC is important for various aspects of prefrontal function, including working memory, affective processing, reasoning, planning, attention, inhibitory control, and the calibration of risk and reward (e.g., [8]). Not surprisingly, these are among the cognitive processes that are still maturing throughout adolescence [9]. In fact, adolescent changes in some cognitive functions appear to occur in parallel to mesocortical dopamine maturation [3,10,11].

Given that prefrontal dopamine fibres continue to increase in density during adolescence, they may be particularly vulnerable to external influences, both positive and negative. Alterations in dopamine connectivity and activity have been linked to cognitive dysfunction observed across various psychiatric conditions [12]. The delayed dopamine connectivity in the PFC during adolescence may underlie, at least in part, the heightened risk of developing mental disorders [13]. Therefore, external influences that can affect adolescent prefrontal dopamine development may have the potential to either exacerbate or protect against the risk of developing certain psychiatric conditions.

Trends

Dopamine axons grow from the nucleus accumbens to the prefrontal cortex during adolescence; this is the only known case of adolescent long-distance axon growth.

The Netrin-1 receptor DCC is responsible for coordinating this growth.

These axons are particularly vulnerable to environmental effects, including recreational drug use, during adolescence and are strategically localised to profoundly influence prefrontal cortex structure and function.

Changes in *DCC* gene expression in humans are linked psychiatric conditions involving prefrontal cortex dysfunction, including schizophrenia and depression.

DCC gene expression is under microRNA control, which may act as a link between environmental factors and prefrontal cortex development.

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To understand both the cognitive changes that occur during adolescence and adolescent vulnerability to mental disorders, it is crucial to clarify the cellular and molecular mechanisms underlying dopamine innervation and connectivity in the PFC. Recent studies using rodent model systems have pointed at the role of specific axon **guidance cues** in this context. In this review, we examine these advances, which lead to a molecular mechanistic explanation for the protracted development of mesocortical dopamine projections. We propose that this emerging picture could explain, at least in part, the particular susceptibility of adolescents to drug addiction and other mental disorders.

The Challenges of Characterising Adolescence across Species

Although we may think of adolescence as a characteristically human attribute, the key behavioural traits associated with the adolescent transition are remarkably consistent across diverse species. Adolescence in humans is characterised by specific, well-described behavioural changes, particularly increased sociality, increased novelty-seeking, and increased risk-taking [14], and these changes are seen in a range of other animal species. The advantage of this is that these organisms can be used to study the neural changes associated with adolescence. However, a challenge to overcome is the need to precisely characterise the developmental time period that constitutes adolescence across multiple species, while keeping in mind that adolescence is a gradual behavioural and cognitive shift that does not have a specific onset or offset.

In the scientific literature, there is a specific developmental timeframe typically referred to as the 'adolescent period'. This is more of an arbitrary standard that encloses adolescence rather than the 'true' adolescent period. In the case of mice and rats, the adolescent period is usually considered to encompass the time from when the animals are weaned until they reach their adult weight [14–19]. In humans, it is usually considered to be between 12 and 19 years of age [14,20]. These adolescent periods are often further divided into three equal thirds constituting 'early', 'mid', and 'late' adolescence. Standardisation of the adolescent period, with all its caveats, allows for reproducible and quantifiable comparisons across species, particularly between model organisms, in our case mice, and humans, whose development we ultimately strive to understand.

Dopamine Development in the Prefrontal Cortex Is Complex and Critical

In the PFC, all cortical layers are innervated by dopamine fibres, which synapse onto pyramidal cells and parvalbumin-expressing interneurons [21–25]. However, the density of fibres varies between layers: density is greater in the superficial and deep cortical layers compared with the intermediate layers [25]. In the rodent, superficial layer dopamine fibres are distinct from deep layer fibres; they are located only in the supragenual (posterior) parts of the cingulate 1 and cingulate 2 subregions and originate from the substantia nigra rather than from the ventral tegmental area (VTA) [2,26]. To our knowledge, studies on the function of this separate yet interesting population of dopamine fibres are lacking. Therefore, we limit our discussion to the better-understood fibres located in the deep layers in the rodent. These fibres are located in the pregenual (anterior) medial PFC and the orbital PFC, and originate from cell bodies located in the VTA.

Dopamine neurons in the VTA, in addition to projecting to cortical regions, also project to subcortical structures, including the nucleus accumbens (NAcc). This **mesocorticolimbic dopamine system** governs key aspects of complex behaviour, including motivation and reward. However, despite their common origin, mesocortical and mesolimbic projections are distinct: they originate from different VTA neuronal populations with specific structures and physiologies [27]. Furthermore, dopamine input to the PFC is sparse compared with the dense input to limbic structures, such as the NAcc [3,28]. Nonetheless, it has been shown that

Glossary

Attentional set: a group of related stimuli that either do or do not contain information relevant to a task, such as acquiring food. By classifying individual stimuli into sets, information processing and decision-making can be performed more efficiently.
Cognitive inhibition: the ability to ignore stimuli that do not contain information relevant to a task, even if the stimuli are relevant to a different task. The Go/No-Go Task is a commonly used test of this ability.
DCC (Deleted in colorectal cancer): acts as a Netrin-1 receptor. It is expressed by growing axons and signals attraction to Netrin-1.
***Dcc* haploinsufficiency:** genetically modified mice that carry one nonfunctional allele of the *Dcc* gene have reduced DCC protein expression in the brain compared with wild-type mice; this is referred to as 'haploinsufficiency'. More refined manipulations can be used to modulate DCC expression in specific populations of neurons and/or time windows, resulting in DCC haploinsufficiency only in specific neuron populations and developmental stages. For instance, the Cre/lox viral-mediated gene-editing technique has been used to generate mice with *Dcc* haploinsufficiency only in ventral tegmental dopamine neurons and from adolescence onwards.
Guidance cue: a protein that signals the appropriate path for growing neurites to follow to their intended innervation target through the developing nervous system.
Mesocorticolimbic dopamine system: a group of brain regions comprising the VTA and forebrain regions that receive its dopamine projections. The 'mesolimbic' component of this system refers to the limbic forebrain regions that receive these projections and includes the NAcc. The 'mesocortical' component of this system refers to the cortical forebrain regions that receive these projections and comprises primarily the PFC.
microRNA: a small RNA that prevents the translation of a specific messenger RNA into protein by binding the 3′-UTR. microRNAs are thought to be important for the 'fine-tuning' of protein expression.

mesocortical dopamine fibres appear to be strategically localised to profoundly influence PFC function [29–31].

Dopamine Connectivity Is still in Flux during Adolescence

Most neuronal growth, maturation, and connectivity occurs during the prenatal and juvenile life stages. By the onset of adolescence, most axonal innervations in the brain have reached their adult densities, including dopamine innervations to limbic regions [32,33]. Although this is true of the NAcc, there is ample evidence that adolescence is a dynamic period in terms of connectivity and receptor expression in this region in particular [34–36]. By contrast, the dopamine innervation to the PFC continues to increase throughout adolescence, achieving final axon density levels during early adulthood [2,3,6,28]. This is accompanied by increases in dopamine synapses onto prefrontal pyramidal neurons [5].

The Unknown Source of the Increasing Mesocortical Dopamine Fibres

It has long been known that the dopamine fibre density in the PFC continues to increase throughout adolescence; however, the source of this increase was unknown. The most parsimonious explanation was that dopamine axons innervating the cortex continue to sprout new branches throughout adolescence. However, an alternate possibility was that additional dopamine axons continue to reach the cortex throughout this period. Such late-stage, long-distance growth would make the dopamine innervation to the PFC all the more unusual.

To study this possibility, a strategy was developed using dual viral injections to label and track axon growth [31]. Using this strategy, dopamine axons were labelled that innervate the NAcc at the onset of adolescence. When these axons were tracked into adulthood, it was found that, although most axons remained in the NAcc, a significant subset grew to the medial PFC [31]. It was also found that, as more dopamine axons innervated the medial PFC, there was a correlated reduction in the axons innervating the NAcc [31]. This was the first clear evidence that protracted mesocortical dopamine development is due to ongoing axonal growth. Given that they grow over such a long distance, mesocortical dopamine axons need to make frequent and cumulative navigational decisions at intermediate targets and likely remain particularly vulnerable to environmental influences throughout this process.

Netrin-1 and its Receptors Coordinate Neural Development throughout Life

What are the molecular mechanisms guiding adolescent mesocortical axon development, and are they linked to the intrinsic maturation of the PFC? During neural development, growing axons find their targets by responding to the coordinated actions of proteins called guidance cues. Guidance cues vary in effective concentration, both spatially across the brain and temporally throughout development, forming signalling pathways that guide growing axons to their intended targets. Notably, the netrin family of guidance cues is found across all bilaterally symmetrical animals [37]. Much work has focussed on Netrin-1's role in mesocortical axon growth and navigation.

Netrin-1 is an approximately 65-kDa protein with three domains [38]. The two N-terminal domains (domains VI and V) are homologues to domains in the laminin protein family, while the C-terminal domain appears unique to netrins and binds to cell surfaces and the extracellular matrix [38,39]. Netrin-1 acts as a guidance cue by influencing the 'decision-making' processes of growing axons, effectively signalling whether, where, and when to grow [39–42]. Netrin-1 also has a critical role in the wiring events that follow axonal path-finding, including target recognition, axon arborisation, and synapse formation, as well as in other processes, such as oligo-axonal adhesion and synaptic plasticity [38,43–45]. The different functions achieved by Netrin-1 appear to be specific to particular developmental periods and maturational states, including adulthood [44,46].

Netrin-1: a guidance cue protein that binds to elements of the extracellular matrix and interacts with receptors expressed on the surfaces of growing axons, signalling axon growth trajectories. Netrin-1 is also heavily involved in other aspects of axon development and maintenance and is expressed in the brain throughout life.

Prefrontal cortex (PFC): the most anterior component of the frontal lobe of the cerebral cortex; defined anatomically by the dense projection it receives from the dorsomedial thalamic nucleus. The PFC is heavily involved in the most complex components of cognition, including planning, reasoning, and decision-making. It receives a sparse dopamine input from the VTA that continues to increase until early adulthood.

Set-shifting: the process of learning that the information contained by different attentional sets has changed. The 'dimension' of an attentional set refers to the sensory modality or property of the set, for example, colour, texture, shape, or odour. Set-shifting may be intra- or extradimensional.

UNC5 (Uncoordinated 5): acts as a Netrin-1 receptor. When bound to Netrin-1, it forms a complex with DCC and signals repulsion from regions of high Netrin-1. Whether the mammalian forms of UNC5 can respond to Netrin-1 in the absence of DCC remains unknown.

Untranslated region (UTR): messenger RNA nucleotide sequences that do not comprise codons and are not translated, but instead contain regulatory sequences. They can be located before the start codon (5′) or after the stop codon (3′).

Netrin-1 organises neuronal circuitries by either attracting or repelling growing axons and dendrites [37]. These opposing responses depend on the binding of Netrin-1 to different receptors or receptor complexes. Two families of Netrin-1 receptors, the **DCC** and the **UNC5 (uncoordinated 5)** families, account for the bifunctional nature of Netrin-1. DCC receptors mediate attraction, whereas UNC5 receptors signal repulsion [47]. These opposing actions can be altered by regulating the availability of DCC and UNC5 receptors at the cell surface [48]. Therefore, the selectivity with which Netrin-1 organises neuronal connectivity can be mediated by the balance between 'attractive' and 'repulsive' receptors expressed at a given time (Figure 1). Subtle alterations in receptor expression can result in selective changes to the organisation of neuronal circuits and, in turn, significant changes to the function of these systems in adulthood. As discussed in the next section, it has been shown that variations in DCC receptor expression during adolescence dramatically influence the development of the rodent mesocorticolimbic dopamine system.

Mapping the Topography of Netrin-1 and DCC in Mesocorticolimbic Dopamine Targets

Dopamine neurons in the VTA highly express DCC receptors throughout life [49,50]. During embryonic rodent development, the onset of DCC receptor expression appears to coincide with dopamine axons starting to grow towards the forebrain [50–52]. DCC continues to be strongly expressed by ventral tegmental neurons throughout postnatal development and, to a lesser degree, in adulthood [50].

In the medial PFC, DCC receptors are expressed by pyramidal neurons [53], but only weakly by dopamine axons [28]. Here, Netrin-1 expression is substantial and localised to pyramidal and parvalbumin-expressing GABA neurons [28], the neurons onto which dopamine axons synapse. Notably, Netrin-1 expression is particularly strong in the deep cortical layers [28], where dopamine innervation is dense. By contrast, in the NAcc, DCC receptors are exclusively expressed by dopamine axons and Netrin-1 expression is much weaker [28]. These complementary expression patterns are maintained before, during, and after adolescence [28]. The high-to-low, receptor-to-ligand expression of DCC and Netrin-1 in dopamine targets suggests an important role for these proteins in the development, organisation, and connectivity of mesocorticolimbic dopamine neurons (Figure 2). We hypothesise that axons expressing copious DCC are sensitive to low concentrations of Netrin-1, while axons expressing minute quantities of DCC respond only to higher concentrations of Netrin-1.

To uncover the actual relationship between Netrin-1 signalling and adolescent dopamine development, researchers have studied the consequences of varying *Dcc* gene expression

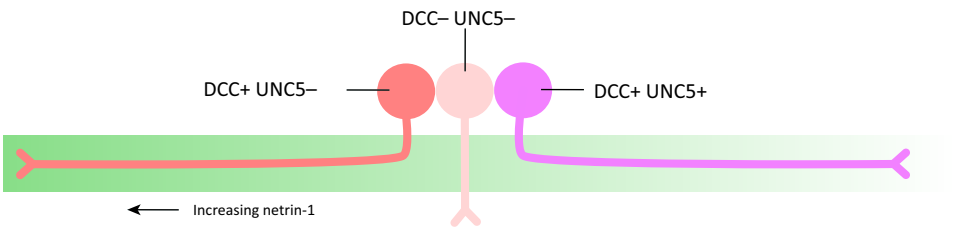


Figure 1. The Responses of Growing Axons to Netrin-1 Gradients. The response of a growing axon to Netrin-1 depends on the expression pattern of Netrin-1 receptors. In general, axons that express deleted in colorectal cancer (DCC) but not uncoordinated 5 (UNC5) (DCC+ UNC5–) are attracted to Netrin-1, whereas axons that express both DCC and UNC5 (DCC+ UNC5+) are repelled by Netrin-1. Axons that express neither DCC nor UNC5 (DCC– UNC5–) do not respond to the Netrin-1 gradient, unless they express other Netrin-1 receptors.

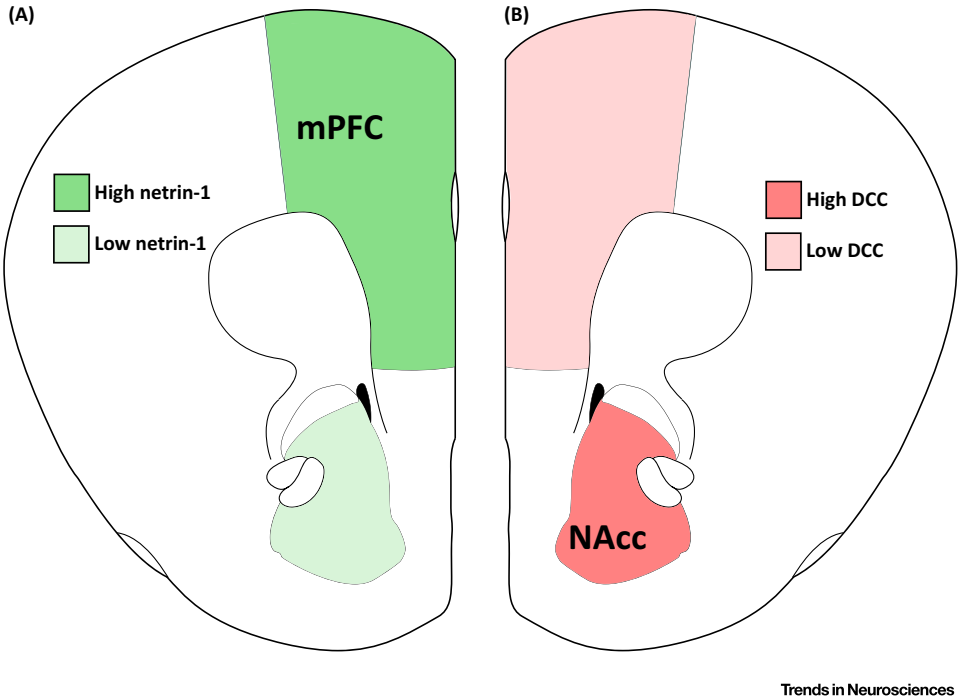


Figure 2. Schematic Illustration of the Complementary Expression Patterns of Netrin-1 and Deleted in Colorectal Cancer (DCC) in Forebrain Dopamine Innervation Targets. The figure depicts a coronal section through the anterior mouse brain. The figure illustrates only the relative levels of DCC expressed by dopamine axons, not DCC expressed by other cell types. (A) Netrin-1 expression is substantial in the medial PFC (mPFC) and much weaker in the nucleus accumbens (NAcc). (B) Dopamine axons in the mPFC rarely express DCC. By contrast, DCC is highly expressed in the NAcc. Note that, in this figure, the frontal 2 cortical region, also referred to as secondary motor cortex, is included in the mPFC. Despite its name, this region is considered to be part of the PFC [86]. Adapted from [87].

with spatial and temporal specificity. The studies utilised *Dcc* heterozygous mice as well as mice with ***Dcc* haploinsufficiency** exclusively in dopamine neurons from adolescence onwards [28,29,31,46]. Given that DCC regulates many processes throughout life, *Dcc* heterozygous mice were used to gain an initial understanding of the role of DCC in adolescence and to develop more specific, testable hypotheses. To confirm the role of the mesocorticolimbic system, all the critical results described here were generated or replicated using mice with targeted *Dcc* haploinsufficiency in dopamine neurons from adolescence onwards. This strategy has uncovered a critical role for DCC signalling in shaping mesocorticolimbic dopamine circuitry. Notably, *Dcc* haploinsufficiency also occurs in humans [54–58].

DCC Receptors Control Dopamine Axon Targeting in Adolescence

As mentioned above, DCC is highly expressed by mesolimbic dopamine axons [28]. By inducing *Dcc* haploinsufficiency specifically in dopamine neurons during adolescence and tracking the growth of their axons, the role of DCC in determining the trajectory of these axons can be studied [31]. Dopamine axons with reduced *Dcc* fail to recognise the NAcc as their intended innervation target and instead reroute to the inner layers of the medial PFC [31]. This results in excess dopamine varicosities and dopamine content in the PFC [28,29,31,49,59]. Given that prefrontal dopamine varicosities are the sites at which most dopamine synthesis, packaging, release, and reuptake occur [60] and because, in the cortex, approximately 93% of dopamine varicosities form synapses [61], it is possible that DCC receptor signalling ultimately controls the number of dopamine synapses in the medial PFC. In summary, DCC receptor

signalling promotes target recognition of mesolimbic dopamine axons in the NAcc, thereby controlling the extent of dopamine innervation to the medial PFC (Figure 3, Key Figure).

DCC-Mediated Wiring of Mesocortical Dopamine Circuitry Determines Complex Cognition in Adulthood

By delimiting mesocortical dopamine input, DCC receptors determine the dendritic arbour complexity, dendritic spine density, and excitability of prefrontal layer V pyramidal neurons [28,29,31,49]. Given these developmental alterations, it was hypothesised that DCC signalling in adolescence influences adult cognitive abilities associated with PFC function, specifically **cognitive inhibition** and behavioural flexibility. To examine cognitive inhibition, mice heterozygous for *Dcc* in dopamine neurons were tested using a Go/No-Go task, in which the animals were trained to respond to a specific cue, as well as to adapt their behavioural responses according to the presence of a secondary cue to obtain a food reward [62]. These mice showed equal performance at all training stages of the task, but during the Go/No-Go phase, showed a specific behavioural alteration; while they had a normal ability to respond during a go trial, they were better able to inhibit responses during a no-go trial [31], indicating that these animals are less impulsive. Improved cognition was also found in *Dcc* heterozygous mice in an attentional **set-shifting** test [63], a test of cognitive flexibility. Cognitive or behavioural flexibility is the ability to make adaptive responses to a changing environment and is specifically tested by the extradimensional set-shifting component of the attentional set-shifting test [29,63,64]. Performance on this component of the test was enhanced in *Dcc* heterozygous mice [29]. By contrast, these mice performed normally in the other stages of the task, including intradimensional set-shifting, indicating that their ability to generate an **attentional set** is not altered [29,63]. By reducing DCC expression on dopamine neurons, it was possible to manipulate the adolescent development of the PFC and selectively alter PFC-dependent behaviours.

By Organising Dopamine Development, DCC Receptors Determine Adult Sensitivity to Stimulant Drugs

Dopamine neurotransmission in the PFC has an important role in regulating dopamine activity in the NAcc. In fact, prefrontal dopamine activity has been shown to attenuate accumbal dopamine release [65]. Given that DCC controls dopamine connectivity and activity in the PFC, it was hypothesised that it may also impact the effects of stimulant drugs that are mediated by dopamine release in the NAcc. These include effects on locomotor activity, conditioned place preference, sensorimotor gating, and sensitivity to intracranial self-stimulation. It was found that, as adults, mice with *Dcc* haploinsufficiency were protected against all these effects [49,59,66–68]. Furthermore, it was recently demonstrated that increased input and concentration of dopamine in the medial PFC directly caused the effects on locomotor activity [69], and we expect that the same is true for the other effects.

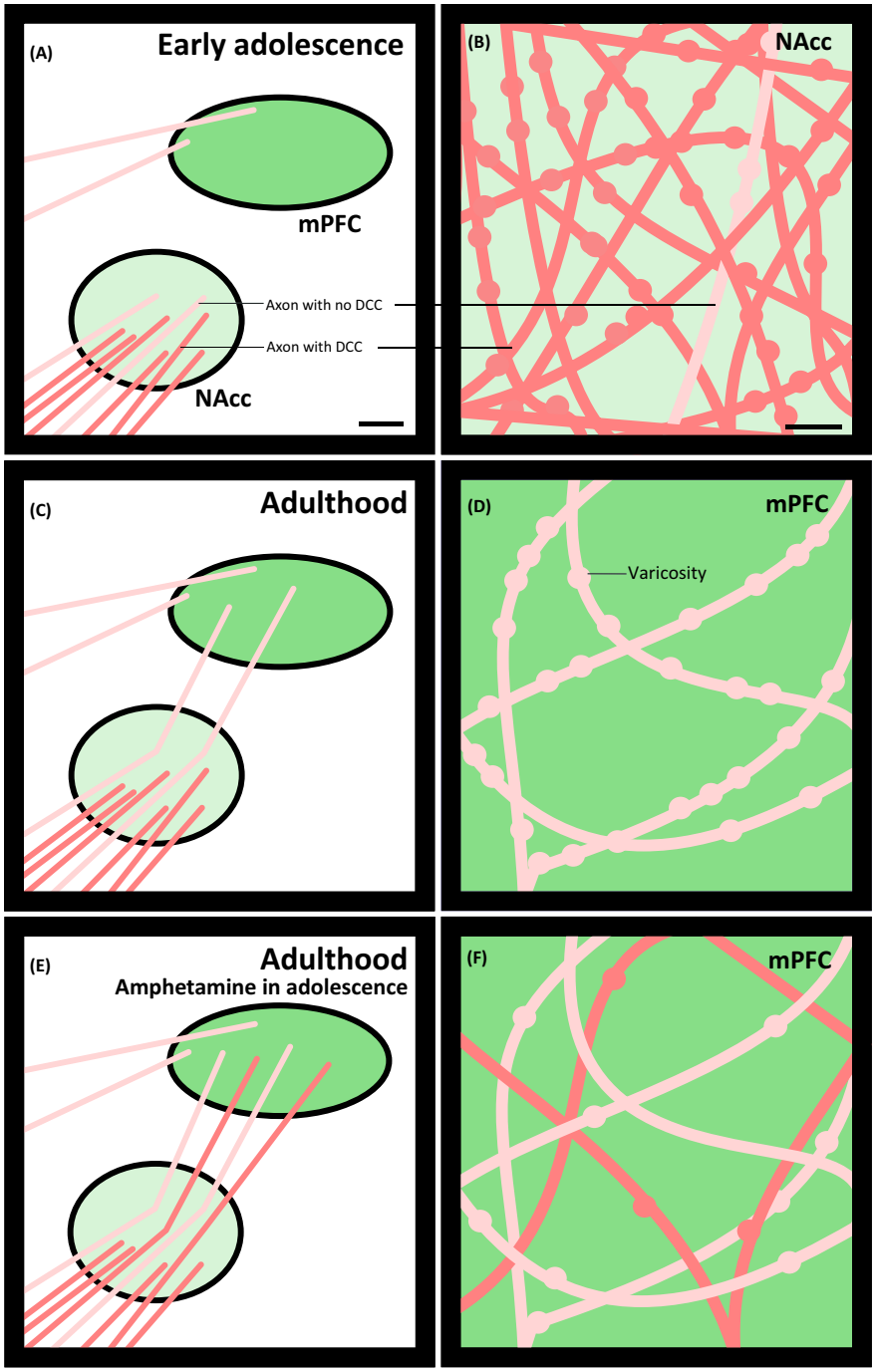
By inducing *Dcc* haploinsufficiency in dopamine neurons in adolescence, dopamine axons are rerouted to the PFC. As noted earlier, this results in a measureable decrease in the dopamine innervation to the NAcc [31]. This reduction may also contribute to the blunted amphetamine-induced dopamine release in the NAcc and, therefore, the blunted behavioural responses.

Addictive Drugs in Adolescence alter DCC Signalling and Dopamine Development

Adolescent exposure to factors that increase the risk of developing disorders involving PFC dysfunction, such as addictive drugs, may cause their effects by altering DCC expression in dopamine neurons. It was shown that repeated exposure to amphetamine in adolescence, but not in adulthood, downregulated DCC receptors in VTA dopamine neurons [70,71]. Remarkably, adult mice that received this treatment during adolescence had an increase in the

Key Figure

Deleted in Colorectal Cancer (DCC) Expression in Dopamine Axons Determines their Forebrain Innervation Target in Adulthood



(See figure legend on the bottom of the next page.)

dopamine input to the PFC [72]. Importantly, the drug regimen used resulted in blood plasma levels in mice comparable to those achieved by adolescent humans who use amphetamine recreationally [73]. As detailed above, reduced DCC expression in adolescence induces target-recognition errors by dopamine axons in the NAcc and their ectopic innervation in the PFC [31]. Therefore, we believe that the increased span in mesocortical innervation observed in amphetamine-treated mice may result from drug-induced target recognition errors in the NAcc and concomitant rerouting to the PFC (Figure 3E,F).

Although amphetamine in adolescence leads to an increase in the expanse of dopamine axons in the medial PFC, it also induces a dramatic reduction in the number and density of their presynaptic sites [72]. This effect is opposite to that observed in transgenic mouse models of *Dcc* haploinsufficiency [28,29]. How amphetamine denudes dopamine axons of presynaptic sites remains to be established. However, aberrant contact and adhesion between dopamine axon terminals and dendrites in the PFC is likely to be involved [74]. This idea is consistent with evidence showing that DCC-mediated Netrin-1 signalling maintains synapses in the PFC [75].

Variations in DCC Receptor Expression Are Linked to Psychiatric Disorders

Dcc haploinsufficiency has been found in several family lineages worldwide [54–58]. There has been great interest in the neuroanatomical and behavioural implications of *Dcc* haploinsufficiency in humans; however, its implications for mesocorticolimbic development and related psychiatric disorders have remained unexamined. Researchers have started to address these questions, and we predict that *Dcc* haploinsufficiency will alter mesocorticolimbic connectivity. We expect that this change will be associated with differences in PFC-dependent cognitive functions, such as cognitive flexibility, reasoning, planning, and reward calibration. We anticipate phenotypic changes to emerge after adolescence. Furthermore, it would be interesting to determine whether subjects with *Dcc* haploinsufficiency show some level of protection against disorders of PFC dysfunction.

Recently, using techniques such as genome-wide association, several studies have suggested links between DCC receptors and psychiatric conditions. One intriguing finding was an association between schizophrenia and polymorphisms in the 3′-**untranslated region** (3′-**UTR**) of the *DCC* gene [76,77]. Although associations between gene polymorphisms and schizophrenia are not uncommon, this finding may be of particular significance because post-transcriptional gene expression regulators modify gene expression by binding the 3′-UTR [78], and therefore such polymorphisms could alter DCC expression levels in dopamine neurons and disrupt mesocorticolimbic system development. A second notable link between psychiatric disease and altered DCC epigenetics is an increase of approximately 50% in DCC gene expression in the PFC of antidepressant-free patients with major depressive disorder who committed suicide [29], a finding later replicated [53]. In fact, there is now substantial

Figure 3. (A) A schematic diagram of the nucleus accumbens (NAcc, light green) and the medial prefrontal cortex (mPFC, dark green) shows that at the onset of adolescence, both dopamine axons that express DCC (red) and ones that do not (pink) innervate the NAcc. Dopamine axons that innervate the mPFC by early adolescence do not pass through the NAcc. Scale bar, ~500 μm. (B) A zoomed-in illustration of the NAcc at the onset of adolescence showing the high density of dopamine input to this region. This level of dopamine innervation remains relatively unchanged throughout adolescence and adulthood. Scale bar, ~10 μm (C) In the adult, dopamine axons that do not express DCC have left the NAcc and now innervate the mPFC. (D) A zoomed-in illustration of the mPFC emphasises that the density of dopamine innervation is lower than in the NAcc. (E) Schematic diagram representing ectopic innervation of dopamine fibres in the adult mPFC following exposure to amphetamine in adolescence. (F) A zoomed-in illustration of the adult mPFC following exposure to amphetamine in adolescence demonstrates that although there is an increase in the number of dopamine axons innervating the mPFC, these axons differ from those seen in nontreated animals; they are denuded of varicosities and some of the innervating axons abnormally express DCC.

correlational genetic evidence from human populations that variation in *DCC* expression is a risk factor for depression [29,53,79–82]. Although such evidence lacks a causative link, using a mouse model of depression, it was recently shown that increased *Dcc* expression in the PFC induced susceptibility to developing depression-like phenotypes [53]. This is the first direct evidence that DCC has a causal role in determining vulnerability to traits that are associated with psychiatric conditions.

The increase in *DCC* expression found in subjects with depression is negatively correlated with the expression of microRNA-218, a potent repressor of DCC expression in human and mouse neurons [53]. **microRNAs** are emerging as molecular links between environmental factors and changes in gene expression [78] and are commonly altered in psychiatric disorders [83]. In general, microRNAs can be relatively easily measured in blood samples [84]; thus, it is feasible that microRNA-218 could be measured in humans. Since circulating levels of microRNAs, including microRNA-218, are known to correlate with levels of microRNAs in the brain, blood-sample measurements of microRNAs, including perhaps microRNA-218, could potentially serve as biomarkers for susceptibility to psychiatric disorders [84,85]. This is a promising strategy for identifying humans, particularly at the onset of adolescence, who may be predisposed to developing psychiatric disorders, including depression. Providing interventions for these individuals before any prodromal symptoms could offer an effective new way to combat adolescent-emergent psychiatric disorders.

Concluding Remarks

Mesocortical dopamine axons represent the only known case of long-distance axon growth during adolescence. As these axons navigate to the PFC, they remain particularly sensitive to environmental influences. The Netrin-1/DCC signalling pathway, which mediates dopamine axon targeting decisions, is a molecular substrate linking environmental events in adolescence to changes in brain maturation and cognition. Notably, animal studies have shown that stimulant drugs acting via Netrin-1/DCC can modify the extent and organisation of the dopamine input to the PFC. Furthermore, changes in DCC expression, including in humans, most likely involving microRNAs, are implicated in psychiatric disorders. Given that microRNAs can be easily measured in blood samples, this presents an opportunity to identify individuals, at a vulnerable age, who may be at risk of developing a psychiatric condition. Contrasting with the detrimental effects of changes in DCC expression, work using transgenic *Dcc* mouse models has also shown the potential of alterations in DCC expression to be protective. Identifying potential environmental influences that can induce these beneficial alterations is an active area of research (see Outstanding Questions). Perhaps the most intriguing of the avenues for further investigation is the possibility of preventing the development of psychiatric conditions in humans through targeted *a priori* interventions. Much further research would be needed to achieve this goal, both to pin down useful biomarkers and to identify effective interventions. Nonetheless, the key contributions of the Netrin-1/DCC pathway to PFC development during adolescence, and, accordingly, its relevance to psychiatric disorders, are becoming clear.

Outstanding Questions

To what extent does adolescent dopamine axon growth contribute to individual differences in vulnerability to psychiatric disorders? Do variations in *DCC* gene expression dictate susceptibility to psychiatric disorders? What is the role of microRNAs?

Does DCC/Netrin-1 signalling have a role in the changes to the endocrine system that coincide with adolescence? If so, is there any interaction between the roles of Netrin/DCC in dopamine and endocrine development?

Do fluctuations in circulating gonadal hormone levels influence the effects of DCC receptors on dopamine axon targeting in adolescence? Do these effects differ between males and females?

Does exposure to other drugs, such as alcohol and marijuana, during adolescence influence the growth of dopamine axons? Does exposure to amphetamine at doses used in therapeutic contexts influence the growth of dopamine axons?

Can resilience to psychopathology be induced through changes in the Netrin-1/DCC signalling pathway? If so, what environmental experiences could trigger these processes?

Is there a role for the other known Netrin-1 receptors [UNC-5, Neogenin, and Down syndrome cell adhesion molecule (DSCAM)] in coordinating dopamine axon growth in adolescence? UNC-5C receptors are expressed in dopamine neurons from adolescence onwards. What are the implications of this for axon targeting?

By controlling the extent of the dopamine input to the PFC, DCC receptors determine the functional organisation of pyramidal neurons. Does the Netrin-1/DCC system also influence the adolescent maturation of GABA interneurons?

Why does late-stage axon growth exist in the brain? Does it have an evolutionary benefit?

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